the algorithm. We had mentioned, though, that OPTICS is a hierarchical clustering algorithm: what does that mean and how does it work with OPTICS?

Hierarchical clustering algorithms produce a multi-layer partitioning of a dataset: this result is often represented with a dendrogram, a tree-like structure that holds a stratified structure; I like to think about exploring dendrograms as an analogous of coring: we can take a section of the dendrogram and check how the flat clustering associated with that section looks like.

But enough with the abstract analogies, let’s delve into an example to clarify how this works!

Figure 12.19 shows a reachability plot obtained from the same reachability distances and ordering as the one in figure 12.17, but with a different value for ε: using a larger value for this radius, obviously the reachability areas around points are larger, and less clusters are formed.

Does this clustering make more sense than the one shown in figure 12.17? Hard to say just by looking at it, answering the question would probably need some domain knowledge… or the tools that we will present in the next section. But since now there is just a single noise point surrounded by three clusters, we might think it could make sense that the points on the left half of the dataset all form a single cluster: to obtain that, we can try a larger value for ε, as shown in figure 12.20.